

Classical ILC/DRP Design from an Algebraic (Matrix-Fraction) Perspective – Part 2: Design

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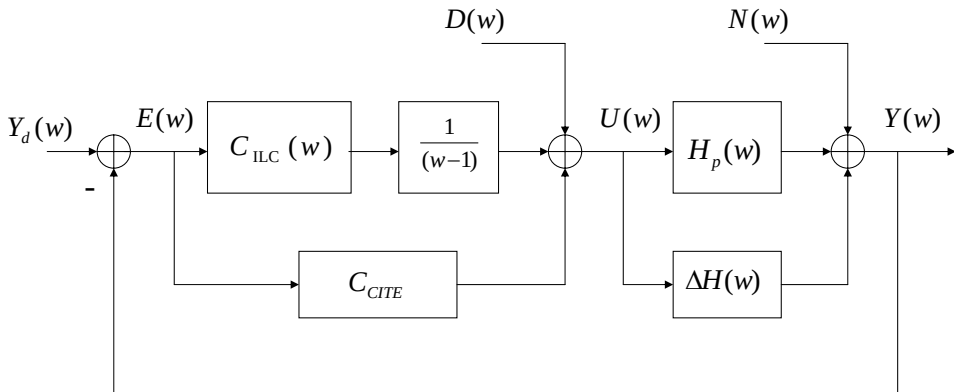
**Pre-Congress Worskhop WS-13 – Learning from Data: Principles, Methods,
and Emerging Ideas & Applications in Iterative Learning Control**

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With Contributions from El-Sharif Omer, Hyo-Sung Ahn, and YangQuan Chen (RIP)



Complete Framework



Categorization of Problems

Y_d	D	N	C	H	
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H_p	Classical Arimoto algorithm
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	$H_p(w)$	Owens' multipass problem
$Y_d(z)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	$N(w)$	$C(w)$	$H_p(w) + \Delta(w)$	Most general problem
$Y_d(z)$	$D(z)$	0	$C(w)$	$H(w) + \Delta(w)$	Frequency-domain uncertainty
$Y_d(z)$	0	$w(t), v(t)$	$\Gamma(w-1)^{-1}$	H_p	Least quadratic ILC
$Y_d(z)$	0	$w(t), v(t)$	$C(w)$	H_p	Stochastic ILC (general)
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H^I	Interval ILC
$Y_d(z)$	$D(z)$	$w(t), v(t)$	$C(w)$	$H(z) + \Delta H(z)$	Time-domain H_∞ problem
$Y_d(z)$	$D(w)$	$w(k, t), v(k, t)$	$C(w)$	$H(w) + \Delta H(w)$	Iteration-domain H_∞ ILC
$Y_d(z)$	0	$w(k, t), v(k, t)$	$\Gamma(k)$	$H_p + \Delta H(k)$	Iteration-varying uncertainty and control
$Y_d(z)$	0	$\tilde{H}$	Γ	H_p	Intermittent measurement problem
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

In This Two-Part Talk ...

- Part 0: Brief Overview of two of these paradigms:
 - ILC
 - DRP
- Part 1: Analysis and design framework for such systems
- **Part 2: Design for discrete ILC/DRP systems**
 - **Monotonic convergence in iteration**
 - Design using current iteration controllers (CITE)
 - Design using optimal PD-type algorithms
 - Iteration-varying gains (not covered today)
 - **Internal model-principle**
 - **Model reference adaptive control (MRAC)**
 - ... and more (not covered today)
 - Robust designs (H_∞ , L_1/l_1 , L_2/l_2 , Interval ILC)
 - ID and Estimation (Kalman filtering in iteration)
 - MPC (norm-optimal - covered in other talks)

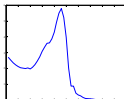


Common Control Structures

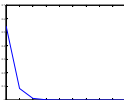
In the remainder of this section we will consider several iteration-domain controllers:

- Two P-type “Arimoto” ILC schemes defined in the literature
 - $u_{k+1}(t) = u_k(t) + k_p e_k(t+1)$
 - $u_{k+1}(t) = u_k(t) + k_p e_k(t)$
- Variations on PD-type ILC schemes:
 - $u_{k+1}(t) = u_k(t) + k_p e_k(t) + k_d (e_k(t+1) - e_k(t))$
 - $u_{k+1}(t) = u_k(t) + \gamma (e_k(t+1) - \beta e_k(t))$
- These PD-type schemes effectively produce a true “D” term along the time axis in the limit as $k \rightarrow \infty$

System 1 does not converge monotonically (in 2-norm):



System 2 does converge monotonically (in 2-norm):



Question: Why do the two systems learn differently?

Framework to Discuss Monotone Convergence (cont.)

- Suppose we have a general higher-order ILC algorithm of the form:

$$u_{k+1}(t) = u_k(t) + L(z)e_k(t+1)$$

where $L(z)$ is a linear (possibly non-causal) filter.

- Then we can represent this ILC update law using supervector notation as:

$$U_{k+1} = U_k + LE_k$$

where L is a Toeplitz matrix of the Markov parameters of $L(z)$.

- For instance, for the Arimoto-type discrete-time ILC algorithm given by

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1)$$

where γ is the constant learning gain, we have $L = \text{diag}(\gamma)$.

Monotonic Convergence Condition

- For the Arimoto-update ILC algorithm, the ILC scheme converges (monotonically) if the induced operator norm satisfies:

$$\|I - \gamma H_p\|_i < 1.$$

- Likewise, a NAS for convergence is:

$$|1 - \gamma h_1| < 1.$$

- Combining these, we can show that for a given gain γ , convergence implies monotonic convergence in the ∞ -norm if

$$|h_1| > \sum_{j=2}^N |h_j|.$$

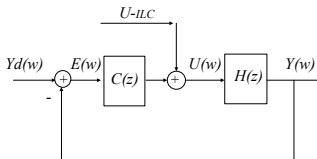
- Note this condition is independent of γ , but instead puts restrictions on the plant.



Control plant to meet convergence condition

Method 1: Current-Cycle Feedback

- Case A:

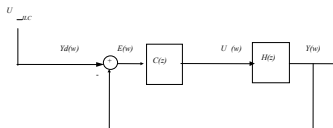


- Plant seen by the ILC algorithm:

$$H_{cl}^A = \frac{H(z)}{1 + C(z)H(z)}.$$

Method 1 (cont.)

- Case B:



- Plant seen by the ILC algorithm:

$$H_{cl}^B = \frac{C(z)H(z)}{1 + C(z)H(z)}.$$

FIR Approach

- Let

$$\begin{aligned} H(z) &= h_1 z^{-1} + h_2 z^{-2} + \cdots, \\ C(z) &= c_0 + c_1 z^{-1} + c_2 z^{-2} + \cdots, \end{aligned}$$

- For Case A the monotonic convergence condition can be shown to be:

$$|h_1| > \sum_{i=2}^N |h_i - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}|.$$

- For Case B the monotonic convergence condition can be shown to be:

$$|c_0 h_1| > \sum_{i=2}^N \left| \sum_{j=1}^i h_j c_{i-j} - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j} \right|.$$

FIR Approach (cont.)

- For both Case A and Case B a controller always exists to give a closed-loop system that satisfies the monotone convergence condition.
- For example, for Case A we can pick:

$$|h_i - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}| = 0,$$

- That is, we solve recursively the following:

$$\begin{aligned} 0 &= |h_2 - h_1 c_0|, \\ 0 &= |h_3 - h_1(h_1 c_1 + h_2 c_0)|, \\ 0 &= |h_4 - h_1(h_1 c_2 + h_2 c_1 + h_3 c_0)|, \\ &\vdots \end{aligned}$$

- Then the system will have monotonic ILC convergence whenever ILC converges.

FIR Approach (cont.)

- Alternately (again for Case A), we can require:

$$|h_i - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}| < \frac{|h_1|}{N-1},$$

- Or, equivalently, we solve the recursive equations:

$$\begin{aligned} \frac{|h_1|}{N-1} &> |h_2 - h_1 c_0| \\ \frac{|h_1|}{N-1} &> |h_3 - h_1(h_1 c_1 + h_2 c_0)| \\ \frac{|h_1|}{N-1} &> |h_4 - h_1(h_1 c_2 + h_2 c_1 + h_3 c_0)| \\ &\vdots \end{aligned}$$

- This approach will be more robust than the previous case.

IIR Approach

- Now, suppose we let $C(z)$ be IIR:

$$H(z) = \frac{n_h(z)}{d_h(z)} = \frac{b_1 z^{-1} + b_2 z^{-2} + \dots + b_n z^{-n}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n}},$$

$$C(z) = \frac{n_c(z)}{d_c(z)} = \frac{\beta_0 + \beta_1 z^{-1} + \beta_2 z^{-2} + \dots + \beta_n z^{-q}}{\alpha_0 + \alpha_1 z^{-1} + \alpha_2 z^{-2} + \dots + \alpha_n z^{-q}}.$$

- Now we have:

Case A:

$$\begin{aligned} H_{cl}^A &= \frac{\Gamma^A(z)}{\Delta(z)} = \frac{n_h(z)}{n_h(z)n_c(z) + d_h(z)d_c(z)}, \\ &= \frac{\gamma_1^A z^{-1} + \dots + \gamma_{(q+n)}^A z^{-(q+n)}}{\delta_0 + \delta_1 z^{-1} + \dots + \delta_{(q+n)} z^{-(q+n)}}, \\ &= h_1^{cl-A} z^{-1} + h_2^{cl-A} z^{-2} + \dots \end{aligned}$$

Case B: Similar.



IIR Approach (cont.)

Define the following vectors:

$$\begin{aligned}
 \mathbf{a} &= (a_0, a_1, a_2, \dots, a_n)^T, \\
 \mathbf{b} &= (0, b_1, b_2, \dots, b_n)^T, \\
 \boldsymbol{\alpha} &= (\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_q)^T, \\
 \boldsymbol{\beta} &= (\beta_0, \beta_1, \beta_2, \dots, \beta_q)^T, \\
 \boldsymbol{\gamma}^A &= (0, \gamma_1^A, \gamma_2^A, \dots, \gamma_{(q+n)}^A)^T, \\
 \boldsymbol{\gamma}^B &= (0, \gamma_1^B, \gamma_2^B, \dots, \gamma_{(q+n)}^B)^T, \\
 \boldsymbol{\delta} &= (\delta_0, \delta_1, \delta_2, \dots, \delta_{(q+n)}^T)^T.
 \end{aligned}$$

Let the appropriately-dimensioned matrices A and B be given as

$$A = \begin{bmatrix} \mathbf{a} & 0 & 0 & \dots & 0 \\ 0 & \mathbf{a} & 0 & \dots & 0 \\ 0 & 0 & \mathbf{a} & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \mathbf{a} \end{bmatrix}, \quad B = \begin{bmatrix} \mathbf{b} & 0 & 0 & \dots & 0 \\ 0 & \mathbf{b} & 0 & \dots & 0 \\ 0 & 0 & \mathbf{b} & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \mathbf{b} \end{bmatrix}$$

IIR Approach

Let

$$H^{cl-A} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ h_1^{cl-A} & 0 & \cdots & \vdots \\ h_2^{cl-A} & h_1^{cl-A} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ h_{q+n+1}^{cl-B} & h_{q+n}^{cl-A} & \cdots & h_1^{cl-A} \\ \vdots & \vdots & \ddots & \vdots \\ h_N^{cl-A} & h_{N-1}^{cl-A} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}.$$

A similar expression can be given for H^{cl-B} .

IIR Approach (cont.)

- Then we can derive

For Case A:

$$\begin{pmatrix} \mathbf{b} \\ 0 \\ \vdots \end{pmatrix} = H^{cl-A}[B|A] \begin{pmatrix} \beta \\ \alpha \end{pmatrix}.$$

For Case B:

$$\begin{pmatrix} \mathbf{B} & 0 \\ 0 & 0 \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} \beta \\ \alpha \end{pmatrix} = H^{cl-A}[B|A] \begin{pmatrix} \beta \\ \alpha \end{pmatrix}.$$



IIR Approach (cont.)

- Hence, given
 - the plant, defined by the Sylvester matrix $[B|A]$ and
 - a desired closed-loop matrix of Markov parameters, H^{cl-A} or H^{cl-B} ,we can solve for the controller, defined by β and α .
- In general the solution of these equations is not known (they are over-determined).
- But, a solution can be possible for high enough controller order, as the null space of $[B|A]$ becomes non-trivial.
- In particular, by forcing the closed-loop system to be deadbeat a solution may be found.

Comments

- With classical Arimoto-type ILC algorithms, the equivalence of ILC convergence with monotonic ILC convergence depends on the characteristics of the plant.
- If a plant does not have the characteristics that ensure such monotonic convergence it is possible to “condition” the plant prior to the application of ILC using current cycle-feedback.
- Two such current-cycle feedback strategies were presented:
 - FIR design (results in high-order controller; always guaranteed, but possible robustness problems)
 - IIR design (solution not always guaranteed)
- Future work will focus on the IIR design approach.

Design for Monotone Convergence

Using the monotonic convergence condition, we have derived ILC algorithm designs using higher-order time-domain filtering to achieve monotonic convergence three ways:

1. Causal current-cycle feedback
2. **Non-causal filtering of the error from the previous trial (optimal design of L for PD-type ILC)**
 - **Examples**
 - **Optimal PD-type ILC Scheme: How to Design**
 - **Optimal PD-type ILC Scheme: Averaged Derivative**
 - **Remarks**
3. Time-varying ILC gain.



Examples: PD-Type ILC

Simulation scenarios:

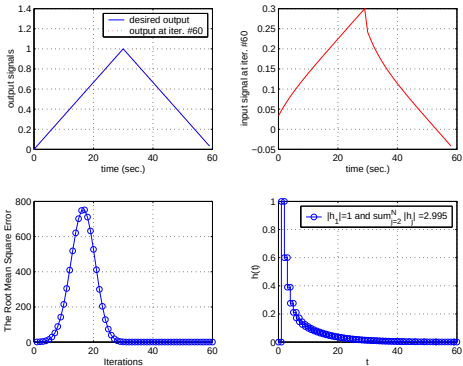
- Second order IIR models are used. All initial conditions are set to 0.
- All plants have $h_1 = 1$, so we fix $\gamma=0.9$ such that $|1 - \gamma h_1| < 1$.
- We fix $N=60$ and max number of iterations = 60.
- The desired trajectory is a triangle given by

$$y_d(t) = \begin{cases} 2t/N & , i = 1, \dots, N/2 \\ 2(N-t)/N & , i = N/2 + 1, \dots, N. \end{cases}$$

- We compare $u_{k+1}(t) = u_k(t) + \gamma e_k(t+1)$ with $u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$

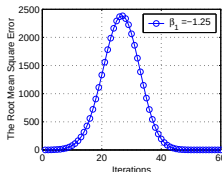
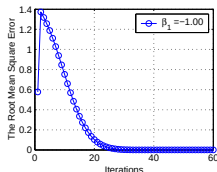
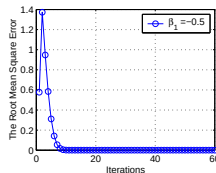
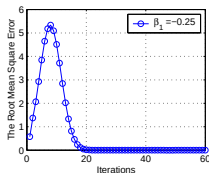
Plant 1a. Stable lightly damped. $H_1(z) = \frac{z-0.8}{(z-0.5)(z-0.9)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1), \gamma = 0.9$$



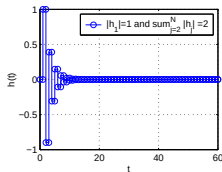
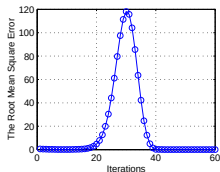
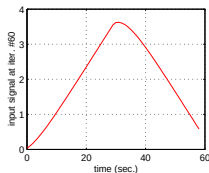
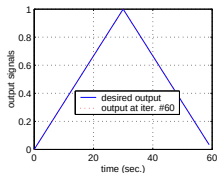
Plant 1b. Stable lightly damped. $H_1(z) = \frac{z-0.8}{(z-0.5)(z-0.9)}$.

$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$ with $\gamma = 0.9$ fixed and β_1 shown on the plots.



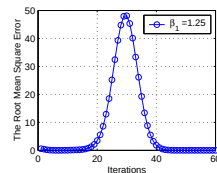
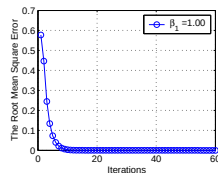
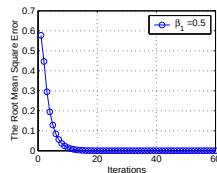
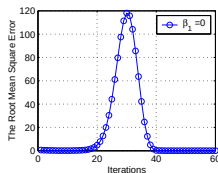
Plant 2a. Stable oscillatory. $H_2(z) = \frac{z-0.8}{(z-0.5)(z+0.6)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1), \gamma = 0.9$$



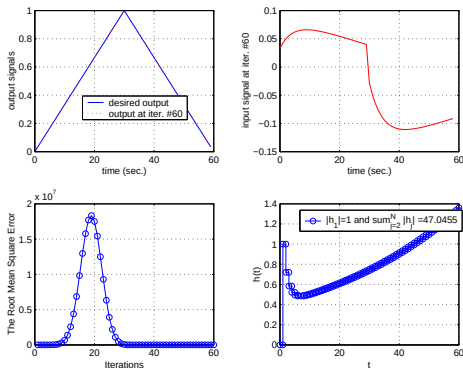
Plant 2b. Stable oscillatory. $H_2(z) = \frac{z-0.8}{(z-0.5)(z+0.6)}$.

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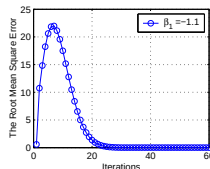
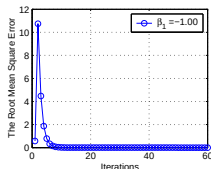
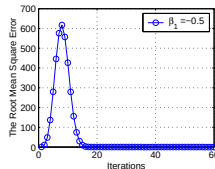
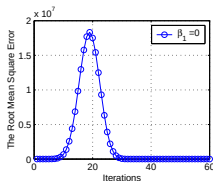
Plant 3a. Slightly unstable. $H_3(z) = \frac{z-0.8}{(z-0.5)(z-1.02)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1), \gamma = 0.9$$



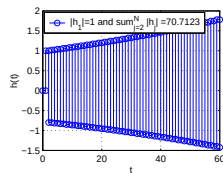
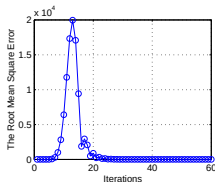
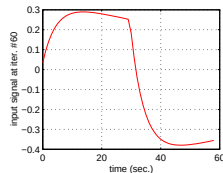
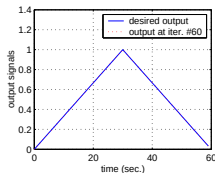
Plant 3b. Slightly unstable. $H_3(z) = \frac{z-0.8}{(z-0.5)(z-1.02)}$.

$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$ with $\gamma = 0.9$ fixed and β_1 shown on the plots.



Plant 4a. Unstable oscillating. $H_4(z) = \frac{z-0.8}{(z+1.01)(z-1.01)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1), \gamma = 0.9$$



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Examples: PD-Type ILC (cont.)

- For $u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$ we conclude that:
 - Monotone convergence is possible for the right values of γ and β .
 - Can relate “overshoot” in convergence for some values of β to zeros in the iteration domain.
- In fact, further, can show:
 - Better convergence behavior is possible with $\beta < 0$.
 - How to pick the optimal β .
- In these simulations we used a simple structure. More generally, we can show how to pick a general lower triangular Toeplitz L (i.e, design of $L(z)$) to find the optimal ILC filter for monotonic convergence.

Comments about Monotonic ILC

- Guaranteeing monotonic convergence of an ILC system is practically important and is theoretically desirable.
- Both higher-order-in-time and first-order-in-iteration have been analyzed with respect to monotonic convergence.
- We found also found that time-varying learning gains could be used for monotonic convergence. This is practically important because without using causal and noncausal bands, the monotonic convergence can be achieved.
- If we just consider the time domain, it is very difficult to guarantee the monotonic condition, while in the iteration domain, the monotonic condition can be achieved relatively easily.
- Various monotonic convergence conditions under various ILC algorithms have been studied. In particular, we have shown that the LMI tool box can be used to design monotonically-convergent ILC algorithms.

Internal Model Principle -1

- Suppose that the reference signal $y_k^d(t)$ has the lifted representation

$$Y_k^d = (y_k(0), y_k(1), \dots, y_k(N-1))^T$$

with a corresponding complex iteration-domain representation

$$Y^d(w) = (Y_0^d(w), Y_1^d(w), \dots, Y_N^d(w))^T$$

- Let $\phi(w)$, a scalar, be the least common denominator of the unstable poles of each $Y_i^d(w)$

Internal Model Principle -2

- Let the denominator of the controller be given as

$$D_c(w) = \phi(w)\bar{D}_c(w)$$



Internal Model Principle -2

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- Applying the final value theorem we can conclude that the steady-state error

$$E_{ss} = \lim_{k \rightarrow \infty} E_{k+1} = \lim_{w \rightarrow 1} (w - 1)E(w) = 0$$

if the zeros of the closed-loop denominator matrix $D_{cl}(w)$ are stable, where

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

Internal Model Principle -2

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- We call this the *Internal Model Principle for ILC/DRPs*



Simultaneous Stability and Tracking -1

- To track an iteration-varying reference signal we need the zeros of

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

to be stable



Simultaneous Stability and Tracking -1

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$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

to be stable

- Note that the plant

$$P(w) = wH^u(wI - H^y)^{-1}$$

is coprime in the space of w -transforms

Simultaneous Stability and Tracking -2

- It follows from standard results that there always exists a stabilizing tracking controller given by

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- We further point out that ILC/DRP control algorithms that do not honor this internal model principle are not robust with respect to tracking performance



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- Closed-loop system stability is defined by the zeros of

$$\begin{aligned} D_{cl}(w) &= (w - 1)(wI - H^y) + (N_1 w + N_0)wH^u \\ &= (I + N_1 H^u)w^2 + (N_0 H^u - I - H^y)w + H^y \end{aligned}$$

Example 1: Constant Reference Signals -2

- In the special case where the control is an Arimoto-style update equation

$$u_{k+1}(t) = u_k(t) + \gamma_1 e_{k+1}(t) + \gamma_0 e_k(t)$$

the gains N_1 and N_0 are diagonal matrices and stability is given by the roots of

$$(1 + \gamma_1 h_0^u)w^2 + (\gamma_0 h_0^u - 1 - h_0^y)w + h_0^y = 0$$

Comments

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Comments

- The resulting controller structure has relative degree zero in the w -domain
- Thus, the controller implicitly involves both past pass information (memory) as well as current pass control
- The claim of arbitrary pole placement is true for the controller shown, except we cannot place closed-loop poles at $w = 0$
 - It is possible to overcome this limitation by making the controller order larger



Simulation -1

- Consider the plant

$$Y_{k+1}(z) = H^u(z)U_{k+1}(z) + H^y(z)Y_k(z)$$

where

$$H^u(z) = \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}}$$
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- Let $y^d(t) = 3 + \sin(0.1t) + 1.5 \sin(0.5(t - 1))$ on the time interval $[1, 30]$



Simulation -2

- Consider two update (or control) laws:
 - ① An internal model controller of the Arimoto form described above

$$u_{k+1}(t) = u_k(t) + 1.5e_k(t) - 0.667e_{k-1}(t)$$

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- ② A unit memory feedback designed for deadbeat convergence

$$u_{k+1}(t) = -\frac{H^y(z)}{H^u(z)}y_k(t) + \frac{1}{H^u(z)}y^d(t)$$

Simulation -3

- To study robustness, after seventy-five iterations the plant is changed to

$$H^u(z) = \frac{1.2632 + 1.3053z^{-1} + 0.0895z^{-2}}{1 + 0.2105z^{-1} - 0.0132z^{-2}}$$
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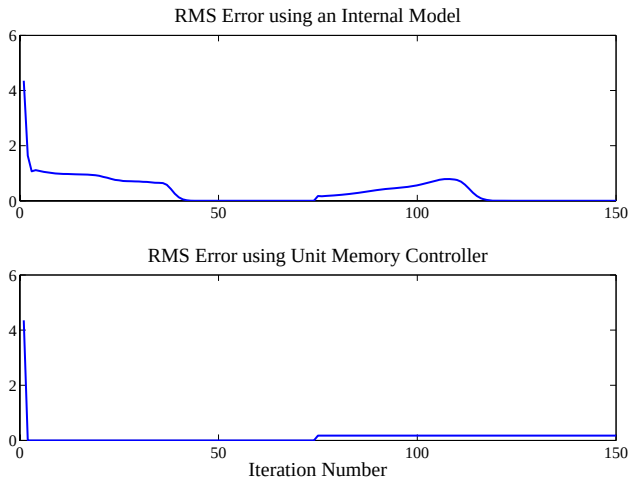
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- Both control laws converge for the nominal plant, but only the internal model-based controller responds properly to the perturbation in the plant

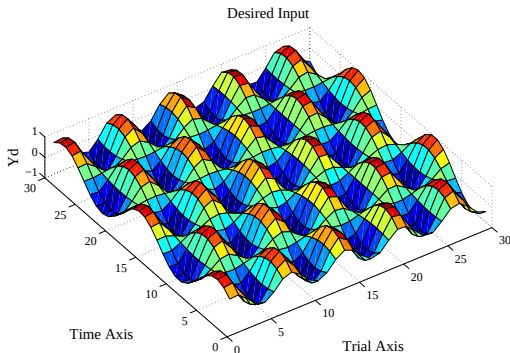
Simulation -4



Example: Sinusoidal Reference -1

- Consider the case when we want to track a signal that varies sinusoidally in iteration

$$y_k^d(t) = \sin(k) \sin(0.4t) \Leftrightarrow Y_t^d(w) = \sin(0.4t) \frac{0.84w}{w^2 - 1.08w + 1}$$



Example: Sinusoidal Reference -2

- Use the update law

$$u_{k+1}(t) = 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - 1.167e_{k-1}(t) + 0.667e_{k-2}(t)$$

which corresponds to the iteration w -domain controller

$$\bar{C}(w) = \frac{1}{w^3 - 1.0806w^2 + w}(N_2w^2 + N_1w + N_0)$$

where $N_2 = I$, $N_1 = \text{diag}(-1.167)$, and $N_0 = \text{diag}(0.667)$



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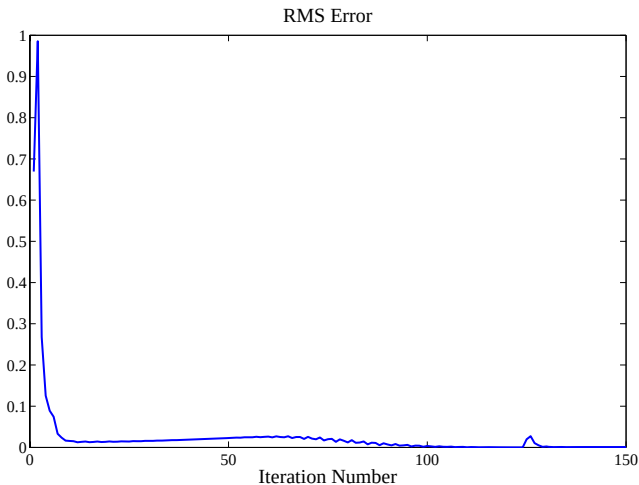
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- These gains give closed-loop poles $w = 0, 0.3403 \pm j0.5909$, chosen to give a quick settling time and minimum overshoot
- Again, the IMP controller gives good robustness to plant perturbations at $k = 125$

Example: Sinusoidal Reference -3



Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$



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- Let the reference model describing the desired behavior of the DRP be $Y_m(k+1) = B_m R(k) + A_m Y_m(k)$ with corresponding transfer matrix of the reference model $G_m(w) = (wI - A_m)^{-1} B_m$



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- We assume only the sign of h_0^u



MRAC Control Law

- We propose the following control law

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- In DRP a solution always exists due to the Toeplitz nature of H^u and H^y



The Adaptive Update Rule

The gains K and L are updated according to

$$\begin{bmatrix} K(k+1) \\ L(k+1) \end{bmatrix} = \begin{bmatrix} K(k) \\ L(k) \end{bmatrix} + \begin{bmatrix} B_m^T P E(k) \text{sign}(l) & 0 \\ 0 & B_m^T P E(k) \text{sign}(l) \end{bmatrix} \Omega(k)$$

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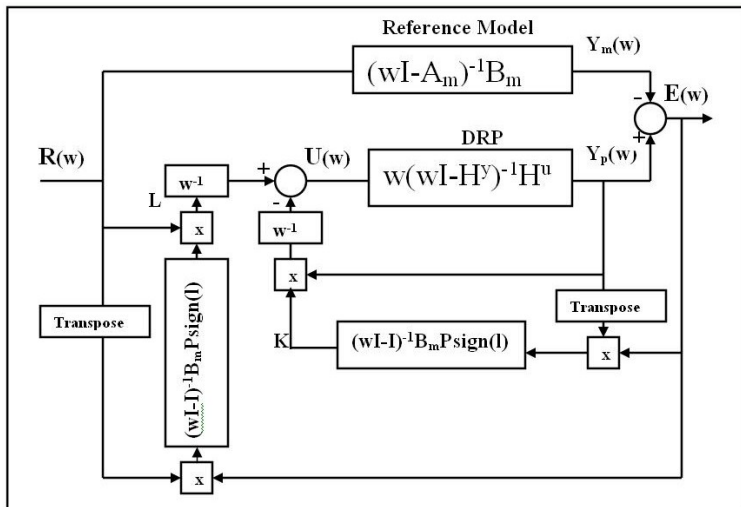
$$\Omega(k) = [Y_p^T(k) \quad -R^T(k)]^T$$

$$\text{sign}(l) = \begin{cases} 1 & \text{if } L^* \text{ is positive definite} \\ -1 & \text{if } L^* \text{ is negative definite} \end{cases}$$

and $P = P^T > 0$ satisfies $A_m^T P A_m - P = -Q Q^T$ for some $Q = Q^T > 0$



Generic MRAC Structure Applied to DRP



MRAC Convergence

- Lyapunov analysis can be used to show
 - The error between the output of the reference model and the output of the plant goes to zero



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- Lyapunov analysis can be used to show
 - The error between the output of the reference model and the output of the plant goes to zero
 - If the signal vector $\Omega(k) = [Y_p^T(k) - R^T(k)]^T$ is persistently exciting then the gains also converge



MRAC Examples

- Consider the same plant as above:

$$Y_p(k+1, z) = H^u(z)U(k+1, z) + H^y(z)Y(k, z)$$

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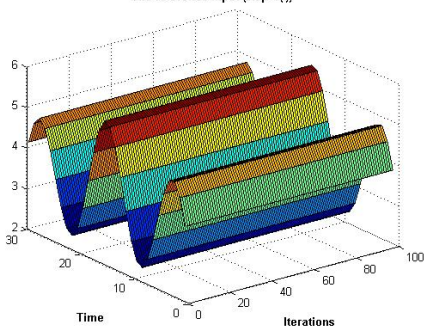
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- Reference model uses $A_m = -0.2I_{30 \times 30}$ and $B_m = 2.0I_{30 \times 30}$

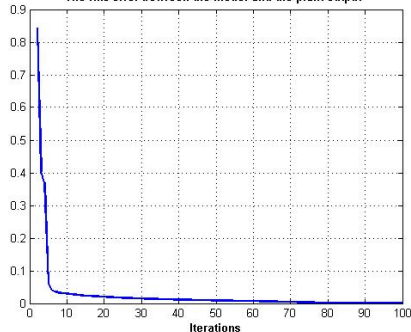


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The reference input (step $r(t)$)

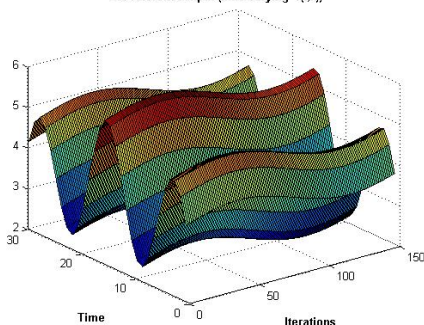


The rms error between the model and the plant output

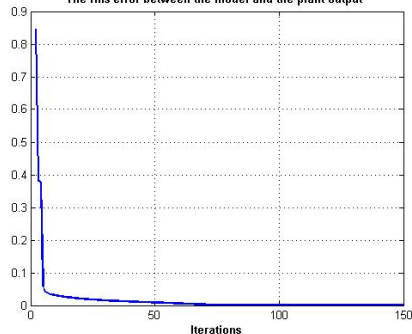


Case 2: $r(t, k) =$
 $3 + \sin(0.1t) + 1.5 \sin(0.5t) + 0.3 \sin(0.04k)$

The reference input (time varying $r(t,k)$)

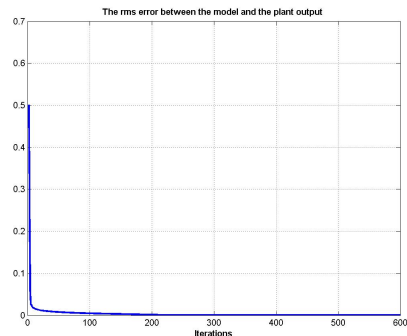
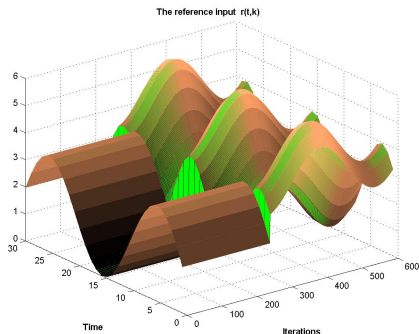


The rms error between the model and the plant output



Case 3: Reference Change After Learning

- Reference is changed after 300 iterations



Algebraic (Matrix Fraction) Approach to ILC/DRP

- Part 0: Brief Overview of two of these paradigms:
 - ILC
 - DRP
- Part 1: Analysis and design framework for such systems
- Part 2: Design for discrete ILC/DRP systems
 - Monotonic convergence in iteration
 - Design using current iteration controllers (CITE)
 - Design using optimal PD-type algorithms
 - Iteration-varying gains (not covered today)
 - Internal model-principle
 - Model reference adaptive control (MRAC)
 - ... and more (not covered today)
 - Robust designs (H_∞ , L_1/l_1 , L_2/l_2 , Interval ILC)
 - ID and Estimation (Kalman filtering in iteration)
 - MPC (norm-optimal - covered in other talks)



Future Work

- Future work can consider the question of poles and zeros in DRPs and how the spectral properties of DRPs can be exploited for MIMO (iteration-domain) controller design



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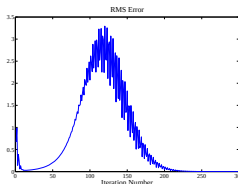
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 - In this case the convergence exhibits the “wave phenomena” well-known in ILC



Further reading on DRP control

- “Internal Model Principle for DRPs,” Kevin L. Moore and El-Sharif Omer, *Proc. 2009 IEEE Conference on Decision and Control*, pp. 446-451, Shanghai, China, December 2009



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- “Norm-Optimal Control of Time-Varying DRPs,” El-Sharif A. Omer and Kevin L. Moore, *Proc. of 2011 IEEE Int. Symp. on Intelligent Control*, Denver, Colorado, Sept 2011
- “Norm-Optimal Control of Time-Varying DRPs with Iteration-Varying Reference Inputs,” El-Sharif A. Omer and Kevin L. Moore, *Proc. of 7th Int. Workshop on Multidimensional (nD) Systems*, Portiers, France, Sept 2011
- “Adaptive Reduced-Order Control of DRPs with Iteration-Varying Reference Signals,” El-Sharif Omer, Kevin L. Moore, *Proc. of 2012 IEEE Int. Symp. on Intelligent Control*, Dubrovnik Palace Hotel, Dubrovnik, Croatia, October 3-5, 2012

Many Problems to Consider

Y_d	D	N	C	H	
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H_p	Classical Arimoto algorithm
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	$H_p(w)$	Owens' multipass problem
$Y_d(z)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
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$Y_d(w)$	$D(w)$	$N(w)$	$C(w)$	$H_p(w) + \Delta(w)$	Most general problem
$Y_d(z)$	$D(z)$	0	$C(w)$	$H(w) + \Delta(w)$	Frequency-domain uncertainty
$Y_d(z)$	0	$w(t), v(t)$	$\Gamma(w-1)^{-1}$	H_p	Least quadratic ILC
$Y_d(z)$	0	$w(t), v(t)$	$C(w)$	H_p	Stochastic ILC (general)
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H^I	Interval ILC
$Y_d(z)$	$D(z)$	$w(t), v(t)$	$C(w)$	$H(z) + \Delta H(z)$	Time-domain H_∞ problem
$Y_d(z)$	$D(w)$	$w(k, t), v(k, t)$	$C(w)$	$H(w) + \Delta H(w)$	Iteration-domain H_∞ ILC
$Y_d(z)$	0	$w(k, t), v(k, t)$	$\Gamma(k)$	$H_p + \Delta H(k)$	Iteration-varying uncertainty and control
$Y_d(z)$	0	$\tilde{H}$	Γ	H_p	Intermittent measurement problem
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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- Thanks for listening!

